

Deploying Water Treatment Energy Flexibility in Power Distribution Systems Operation

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Abstract—As the global demand for freshwater resources continues to grow at a fast pace, energy costs associated with treating and distributing water in urban areas are projected to increase. A promising solution to enhance the efficiency and reduce these costs is to coordinate the operation of water systems with power distribution systems operation. In this paper, we propose an optimization framework which integrates the energy flexibility of water treatment, desalination and distribution systems in power distribution systems operation. In the proposed framework, a water distribution system operator (W-DSO) co-optimizes the operation of variable speed pumps and water storage tanks in water treatment, desalination and distribution systems for minimizing its daily energy costs, given the expected water demand and energy tariffs. The optimized energy flexibility of W-DSO is then incorporated in the operation of power distribution systems. The proposed framework takes into account the operational constraints of both power and water distribution networks, thus co-optimizing their operation without endangering the reliable supply of power and water to customers. Simulation results demonstrate the merits of the proposed model in enhancing the economic efficiency of the test 15-node water network supplied by a 33-bus power distribution system.

Index Terms—Energy-water nexus, energy flexibility, water treatment, desalination, power distribution systems

I. INTRODUCTION

INTERDEPENDENCE between power and water infrastructures has recently been in the spotlight for both power and water system operators who seek to better understand how a synergistic operation between their systems could improve energy efficiency and reduce their operating costs. This synergy is commonly referred to as the *energy-water nexus* in the sense that the use of one system depends on the availability of the other [1]. Water is needed in many stages of power generation, from thermal power plant cooling to fossil fuel extraction, while power is required to treat and distribute large amounts of water over long distances [2]. As increasing pressures of drought and population growth arise, the interdependence between power and water distribution systems will be intensified.

Traditionally, power and water distribution systems have been planned, designed, and operated with the assumption that energy and water resources were, and would continue to be, inexpensive and abundant [3]. However, current and upcoming challenges related to resource availability and operating expenses have forced power and water distribution systems operators to reconsider their operation strategies, whereby sustainability and cost savings have become core values [4].

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The interdependence between the main components of power and water distribution systems is schematically shown in Fig. 1. In Fig. 1, the power distribution system supplies the electricity demand of the water distribution system, which consists of two main subsystems: water treatment subsystem and water conveyance subsystem. The water treatment subsystem, which includes wastewater treatment and desalination plants, removes contaminants and salt from water before being delivered to end users via the water conveyance subsystem. These subsystems consist of variable speed pumps (VSPs), water storage tanks, and desalination plants. The VSPs consume electricity to generate sufficient pressure in water pipes needed to convey water over long distances or within the water treatment plant, in addition to storing freshwater in the storage tanks. Further, VSPs can rapidly adjust their speed (therefore their power consumption) to balance water load fluctuations [5]. As a result, pumping loads can be considered as dispatchable loads in power distribution systems.

Water storage tanks are indispensable components of the water conveyance subsystem that store or release water respectively during periods of low or high freshwater demand. As the pumping energy required to deliver water to storage tanks is flexible via the utilization of VSPs, water pumping schedule into tanks could be optimized to provide flexibility to power distribution systems [6], [7]. Water filtration membranes are part of the water treatment subsystem, which purifies water for human consumption [8], [9]. In order to ensure freshwater productivity, VSPs are utilized on the feed side of membranes to pressurize brackish or saline water. Considering the flexible operation of VSPs, water treatment plants are also candidates to provide flexibility in power systems.

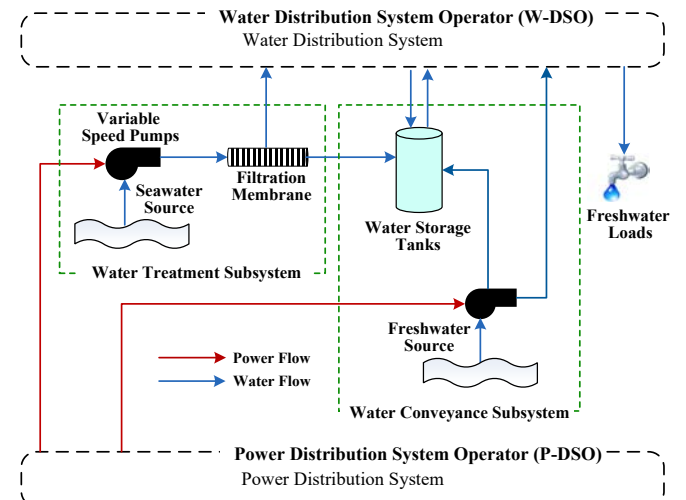


Fig. 1. Power and water distribution systems interdependence

Given the tight interdependence between power and water distribution systems, a coordinated operation could uncover significant operational benefits for both infrastructures. In addition, the flexible water components, if properly managed, could reduce operating costs in both power and water systems, and provide a range of services (e.g., flexible energy, frequency regulation) to the power grid [10]–[12]. In this context, this paper proposes a framework to integrate the energy flexibility of water distribution, treatment and desalination systems in power distribution systems operation. More specifically, a day-ahead WDS operation model is proposed that co-optimizes the operation of VSPs, tanks, and treatment plants for minimizing the operation cost of WDS, given the expected water demand and energy tariff of the next day. The solution of the proposed WDS operation problem provides the optimized energy consumption schedule for VSPs, which aggregated energy consumption accounts for the energy flexibility capacity of the WDS that is offered to the power distribution systems operator (P-DSO). The P-DSO, then, solves a proposed optimal power flow problem for minimizing the day-ahead operation cost of its power distribution system, integrating the energy flexibility of WDS. The proposed framework incorporates detailed formulations of the water and power flow constraints in order to ensure that the delivery and reliance on WDS energy flexibility would not deteriorate the operational performance of both water and power distribution systems.

II. WATER DISTRIBUTION SYSTEM OPERATION MODEL

The WDS delivers freshwater from the freshwater reservoirs to the water consumers, as well as seawater from the seawater reservoirs to the water treatment plants via a system of pipes, water storage tanks, and variable speed pumps. The objective function of the W-DSO operation model is to minimize the operation cost for meeting the freshwater demand of WDS over the scheduling horizon of NT time intervals, as follows:

$$\min \sum_t^{NT} (\mathbf{1}_U^T \mathbf{P}_t^U \rho_t^e) + \sum_t^{NT} (\mathbf{M}_t \Theta^F \mathbf{W}_t^F + \Xi_t \Theta^E \mathbf{W}_t^E). \quad (1)$$

The first term in (1) is the pumping energy cost, where $\mathbf{P}_t^U$ is the $U \times 1$ decision variable vector of pumps power consumption; ρ_t^e is the forecasted WDS energy tariff at time t that the W-DSO pays to the P-DSO to consume energy; and $\mathbf{1}^U$ is a $U \times 1$ vector of ones. The second term in (1) represents the cost incurred for purchasing water from the freshwater reservoirs, where $\mathbf{M}_t$ is $1 \times F$ vector of freshwater price at time t ; Θ^F is $F \times N$ freshwater reservoir-node incidence matrix; and $\mathbf{W}_t^F$ is $N \times 1$ decision variable vector of freshwater inflows at freshwater reservoir nodes at time t . The third term in (1) represents costs related to the seawater pretreatment process, where Ξ_t is $1 \times E$ vector of pretreatment seawater price at time t ; Θ^E is $E \times N$ seawater reservoir-node incidence matrix; and $\mathbf{W}_t^E$ is $N \times 1$ decision variable vector of seawater inflows in seawater reservoir nodes at time t .

The W-DSO operation model is subjected to the following water distribution, treatment and desalination constraints:

$$\mathbf{W}_t^E + \mathbf{W}_t^F - \mathbf{W}_t^D = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda}^T [\mathbf{C}_t - \mathbf{D}_t], \quad \forall t, \quad (2)$$

$$0 \leq \mathbf{W}_t^F \leq \overline{\mathbf{W}}^F, \quad \forall t, \quad (3)$$

$$0 \leq \mathbf{W}_t^E \leq \overline{\mathbf{W}}^E, \quad \forall t, \quad (4)$$

$$\mathbf{1}^N \mathbf{Y}_t^T \mathbf{\Lambda} \mathbf{A}^T \mathbf{W}_t^E \geq \alpha \mathbf{1}^N \mathbf{W}_t^D, \quad \forall t, \quad (5)$$

$$-\mathbf{A}^T \mathbf{H}_t = \mathbf{G} \Pi(\mathbf{Q}_t) |\mathbf{Q}_t|^{1.852}, \quad \forall t, \quad (6)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_t \leq \overline{\mathbf{H}}, \quad \forall t, \quad (7)$$

$$\Theta^F \mathbf{H}_t = \mathbf{Z}^F, \quad \forall t, \quad (8)$$

$$\Theta^E \mathbf{H}_t = \mathbf{Z}^E, \quad \forall t, \quad (9)$$

$$-\overline{\mathbf{Q}} \leq \mathbf{Q}_t \leq \overline{\mathbf{Q}}, \quad \forall t, \quad (10)$$

$$\mathbf{Q}_t^C = \mathbf{L}^T \mathbf{K} (\Delta \mathbf{H}_t - \Delta \pi_t), \quad \forall t, \quad (11)$$

$$\Delta \mathbf{H}_t = \mathbf{\Lambda} \mathbf{A}^T (\mathbf{H}_t^S - \mathbf{H}_t^C), \quad \forall t, \quad (12)$$

$$\Delta \pi_t = \Pi(\mathbf{1}^O \odot (\mathbf{1}^O - \mathbf{Y}_t)), \quad \forall t, \quad (13)$$

$$\mathbf{Y}_t = \mathbf{L} (\mathbf{Q}_t^C \odot \mathbf{Q}_t^S), \quad \forall t, \quad (14)$$

$$\Delta \mathbf{H}_t \geq \Delta \pi_t, \quad \forall t, \quad (15)$$

$$\mathbf{V}_{t+1} = \mathbf{V}_t + \mathbf{C}_t - \mathbf{D}_t, \quad \forall t, \quad (16)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_t \leq \overline{\mathbf{V}}, \quad \forall t, \quad (17)$$

$$\mathbf{V}_1 = \mathbf{V}^{init}, \quad (18)$$

$$\mathbf{V}_t = \Delta \mathbf{\Lambda} \mathbf{H}_t, \quad \forall t, \quad (19)$$

$$0 \leq \mathbf{C}_t \leq \overline{\mathbf{C}}, \quad \forall t, \quad (20)$$

$$0 \leq \mathbf{D}_t \leq \overline{\mathbf{D}}, \quad \forall t, \quad (21)$$

$$\mathbf{B}^U \mathbf{A}^T \mathbf{H}_t = \Omega_t^{U^2} \odot \left(a \mathbf{1}^{U-b} \left((\mathbf{1}^U \odot \Omega_t^U) \odot \mathbf{B}^U \mathbf{Q}_t \right)^c \right), \quad \forall t, \quad (22)$$

$$\mathbf{P}_t^U = \Omega_t^{U^3} \odot \left(d \mathbf{1}^{U-e} \left((\mathbf{1}^U \odot \Omega_t^U) \odot \mathbf{B}^U \mathbf{Q}_t \right) \right), \quad \forall t, \quad (23)$$

$$0 \leq \mathbf{P}_t^U \leq \overline{\mathbf{P}}^U, \quad \forall t, \quad (24)$$

$$\mathbf{B}^U \mathbf{Q}_t \geq 0, \quad \forall t. \quad (25)$$

The water balance constraint is given in (2), where $\mathbf{W}_t^D$ is $N \times 1$ vector of nodes' freshwater demand parameter at time t ; $\mathbf{Q}_t$ is $J \times 1$ decision variable vector of volumetric flow rate of pipes at time t ; $\mathbf{C}_t$ and $\mathbf{D}_t$ are respectively $T \times 1$ vectors of freshwater inflow and outflow decision variables of water tanks at time t ; $\mathbf{\Lambda}$ is $T \times N$ tank-node incidence matrix; and $\mathbf{A}$ is $N \times J$ node-arc incidence matrix. The freshwater and seawater inflows are respectively limited in (3) and (4), where $\overline{\mathbf{W}}^F$ and $\overline{\mathbf{W}}^E$ are the maximum freshwater and seawater inflow bounds. Based on the availability of freshwater resources and the possibility of facing freshwater shortages, W-DSO may recover higher portions of freshwater from the desalination plants. This is mathematically expressed in (5), where $\mathbf{Y}_t \in [0, 1]$ is $O \times 1$ vector of the membranes' freshwater recovery ratio at time t ; $\mathbf{L}$ is $O \times J$ membrane-arc incidence matrix; $\mathbf{1}^N$ is a $1 \times N$ vector of ones; and $\alpha \in [0, 1]$ is the freshwater unavailability index that remains fixed throughout the scheduling horizon. The head loss in pipes is approximated by the empirical *Hazen-Williams* constraint (6), where $\mathbf{H}_t$ is $N \times 1$ decision variable vector of nodes' pressure head at time t ; $\mathbf{G}$ is $J \times J$ diagonal matrix of pipes' resistance parameters, and $\Pi(\cdot)$ is unit sign function [13]. The nodal pressure heads are confined in (7) to the minimum and maximum pressure head bounds at the nodes. The nodal pressure heads are respectively fixed at the freshwater and seawater reservoir nodes in (8) and (9) by a fixed elevation height, where $\mathbf{Z}^F$ and $\mathbf{Z}^E$ are respectively $F \times 1$ and $E \times 1$ vectors that contain freshwater

and seawater reservoir nodes' elevation data. The volumetric flow rate variable of arcs is limited in (10), where the bound $\bar{Q}$ depends on the pipes' and pumps' design characteristics.

The pressures and flow rates for the RO membrane are presented in (11)-(15). The permeate flow rate $Q_t^C \subset Q_t$ is approximated in (11), where ΔH_t is $O \times 1$ vector of pressure difference across the membrane, also referred to as transmembrane pressure; $\Delta \pi_t$ is $O \times 1$ vector of the average osmotic pressure difference between the retentate and permeate streams along the membrane; K is $O \times O$ diagonal matrix of membranes' active surface area and hydraulic permeability parameters. In (12), the pressure difference across the membrane is equal to the difference between the pressure of the feed water at the entrance of the membrane $H_t^S \subset H_t$ and the pressure of the permeate water at the exit of the membrane $H_t^C \subset H_t$. The average osmotic pressure difference is approximated in (13), where the membrane's freshwater recovery ratio Y_t is equal to the ratio of permeate water flow rate to the feed water flow rate $Q_t^S \subset Q_t$ in (14); Π is $O \times O$ diagonal matrix of the osmotic pressure coefficients of the feed water.

The state equation that describes the dynamic operation of water storage tanks is given in (16), where V_t denotes $T \times 1$ vector of water volume in tanks at time t . The water volume of tanks is confined to the minimum and maximum volume limits in (17). The water volume stored in tanks at the beginning of scheduling horizon is initialized in (18), where V^{init} is the vector of initial water volume stored in tanks. In (19), the water volume varies based on the pressure head at the tank nodes, where Δ is $T \times T$ diagonal matrix containing tanks' surface values. Constraints (20) and (21) respectively limit tanks' water inflow and outflow rates to the maximum inflow and outflow rate bounds. The raise of pressure head in pipes generated by variable speed pumps and their power consumption are respectively given in (22) and (23), where Ω_t^U is $U \times 1$ decision variable vector of pumps speed at time t ; B^U is $U \times J$ pump-arc incidence matrix; and a, b, c, d and e are pump-specific parameters [14]. Signs $\odot$ and $\oslash$ in (22) and (23) refer respectively to the component-wise matrix multiplication and division. In (24), $U \times 1$ decision variable vector of pumps power consumption, P_t^U , is constrained to its maximum power capacity, P^U , while the water flow rate through a pump takes positive values in (25), denoting that pumps can operate only in one direction.

III. POWER DISTRIBUTION SYSTEM OPERATION MODEL

In the proposed model, the P-DSO integrates the energy flexibility of WDS in its operation by utilizing the proposed optimal power flow model. The proposed model integrates the branch-flow model for modeling power distribution networks [15], which accounts for both active and reactive power flows and integrates the active and reactive power consumption flexibility of VSPs in power distribution buses. The objective function of proposed model is to minimize the operation cost for supplying the electricity demand of power distribution system over scheduling horizon of NT intervals:

$$\min \sum_t^{NT} (P_t^G \lambda_t^e), \quad (26)$$

where P_t^G is the active power flowing from the upstream transmission system to supply the distribution system at time t and λ_t^e is the forecasted wholesale energy price at time t . The proposed model is subjected the following constraints:

$$P_t^G = \sum_{1k \in \mathcal{L}} P_{1k,t}^L + g_1 V_{1,t}^{sq}, \quad \forall t, \quad (27)$$

$$Q_t^G = \sum_{1k \in \mathcal{L}} Q_{1k,t}^L + b_1 V_{1,t}^{sq}, \quad \forall t, \quad (28)$$

$$-P_{i,t}^{D,I} - \sum_{u \in \mathcal{U}_i} P_{u,i,t}^U = \sum_{ik \in \mathcal{L}} P_{ik,t}^L - \sum_{ji \in \mathcal{L}} (P_{ji,t}^L - r_{ji} I_{ji,t}^{sq}) + g_i V_{i,t}^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (29)$$

$$-Q_{i,t}^{D,I} - \sum_{u \in \mathcal{U}} Q_{u,i,t}^U = \sum_{ik \in \mathcal{L}} Q_{ik,t}^L - \sum_{ji \in \mathcal{L}} (Q_{ji,t}^L - x_{ji} I_{ji,t}^{sq}) + b_i V_{i,t}^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (30)$$

$$V_{i,t}^{sq} - V_{j,t}^{sq} = -2(r_{ji} P_{ji,t}^L + x_{ji} Q_{ji,t}^L) + (r_{ji}^2 + x_{ji}^2) I_{ji,t}^{sq}, \quad (ji) \in \mathcal{L}, \forall t, \quad (31)$$

$$\underline{V}_i^{sq} \leq V_{i,t}^{sq} \leq \bar{V}_i^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (32)$$

$$V_{i,t}^{sq} I_{ji,t}^{sq} \geq P_{ji,t}^2 + Q_{ji,t}^2, \quad (ji) \in \mathcal{L}, \forall t, \quad (33)$$

$$I_{ji,t}^{sq} \leq \bar{I}_{ji}^2, \quad (ji) \in \mathcal{L}, \forall t. \quad (34)$$

The active and reactive power flows in the substation bus, $i = 1 \in \mathcal{B} = \{1 \dots B\}$, are formulated in (27) and (28), where $(P_{ji,t}^L + iQ_{ji,t}^L)$ is the complex power flow in line $ji \in \mathcal{L}$ at time t ; The active and reactive power balance constraints in distribution buses, $i \in \mathcal{B}^b = \{2 \dots B\}$, that integrate the WDS flexible pumping loads are respectively formulated in (29) and (30), where $(P_{u,i,t}^U + iQ_{u,i,t}^U)$ is the complex power consumption of a VSP $u \in \mathcal{U}$ connected to distribution bus $i \in \mathcal{B}^b$ at time t ; $V_{i,t}^{sq}$ is the squared voltage magnitude at bus $i \in \mathcal{B}$; and $I_{ji,t}^{sq}$ is the squared current magnitude of line $ji \in \mathcal{L}$ at time t . The voltage drop constraint is formulated in (31) in terms of the active and reactive power flows of the lines, where $z_{ji} = r_{ji} + ix_{ji}$ is the complex impedance. The bus voltage limits are enforced in (32). Constraint (33) limits the complex power flows of the lines, while the current flows are constrained to the maximum current limit of lines in (34).

IV. SIMULATION RESULTS

The proposed model is implemented in a layout where the 33-bus power distribution test system [16] supplies the electricity of four 15-node water distribution test systems [17] connected to buses 16, 22, 25, and 33. The topology of the power and water distribution systems are respectively shown in Fig. 2 and Fig. 3. In Fig. 2, the electric demand of the system is served by the upstream power transmission network, as well as from four distributed solar generating (DSG) units connected to buses 16, 22, 23, and 30 through controllable inverters. The CAISO hourly load data for March 30, 2018 is scaled down to the test system's active and reactive peak loads of 3,715kW and 2,300kVAr and used in the simulations. The CAISO's daily active solar power profile of the same day is scaled down to the inverter's rated active power output of 225kW. In Fig.

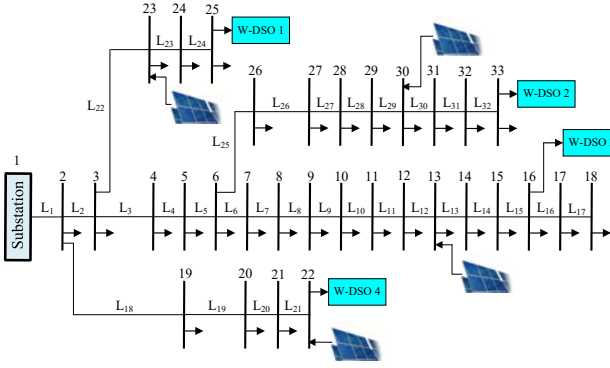


Fig. 2. 33-Bus power distribution test system

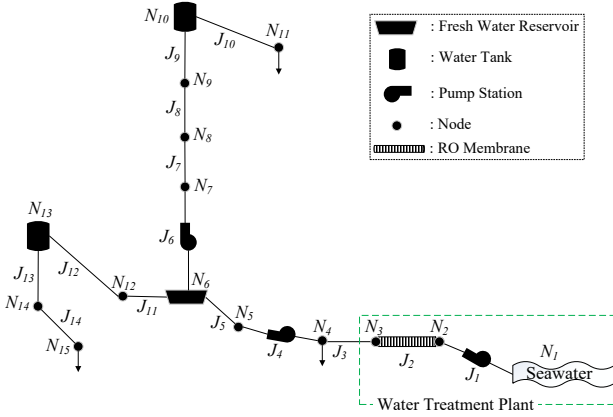


Fig. 3. 15-Node water distribution test system

3, the WDS consists of 15 nodes, 14 arcs (including 10 pipes, 3 VSPs in arcs 1, 4, 6, and 1 RO membrane in arc 2), 2 water tanks (nodes 10, 13), 3 water load nodes (nodes 4, 11, 15), 1 freshwater reservoir (node 6), and 1 seawater reservoir (node 1). Further, we assume that the freshwater unavailability index, α , is 0.5, implying that at least 50% of the total freshwater demand is supplied by the treatment plant. The freshwater recovery ratio, Y_t , of the water treatment plant is set to vary between 0 and 0.5.

The numerical results are provided for three cases aiming at investigating how different energy tariff schemes could impact the energy flexibility of the WDS as well as the operation costs of both power and water distribution systems. In case 1, the W-DSO optimizes the operation of its WDS by taking advantage of the low WDS energy tariffs during night hours. In Case 2, the WDS energy tariff remains constant during the scheduling horizon, implying that the P-DSO provides no incentives to the W-DSO for changing its operation schedule. In Case 3, the WDS energy tariff follows the trend of the wholesale energy price that significantly drops during high solar availability hours. The WDS energy tariffs for the three cases are shown in Fig. 4. In all cases, the energy consumption of four WDSs is optimized using the model proposed in Section II, and WDS energy flexibility is integrated in power distribution systems operation using the model proposed in Section III.

1) *Daily Operation Costs*: The daily energy costs of P-DSO and W-DSO in test power and water distribution systems are shown in Table I. In Table I, the total energy cost in Case 2

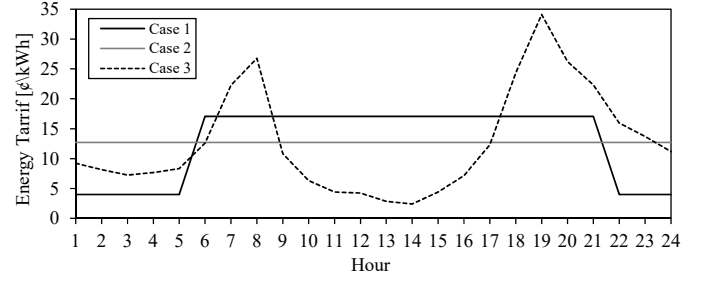


Fig. 4. Energy tariff applied to water distribution system

is the highest among all cases, \$3,504.6, showing that the flat energy tariff scheme significantly limits WDS energy flexibility. In Case 1, the total operation cost is reduced to \$3,400.6 as compared to Case 2, demonstrating the operational benefits, which result from shifting pumping loads to night hours where energy tariffs are low. The total operation cost in Case 3 is further reduced to \$3,323.4 as compared to Case 2, showing that the dynamic WDS energy tariff scheme in Case 3 leads to greater operational benefits for both systems as a result of increasing the WDS energy flexibility capacity.

TABLE I
DAILY ENERGY COSTS

	P-DSO Cost (\$)	W-DSO Cost (\$)	Total Cost (\$)
Case 1	2,839.9	560.6	3,400.6
Case 2	2,902.2	602.3	3,504.5
Case 3	2,780.1	543.2	3,323.4

2) *Power Consumption Schedules*: The hourly active power consumption schedules of the water and power distribution systems are respectively displayed in Fig. 5 and Fig. 6. In Fig. 5, the active power consumption schedules of the WDS in three cases follow the profiles of the WDS energy tariffs, illustrated in Fig. 4. More specifically, in Case 1, the W-DSO operates the pumps to leverage the low energy tariffs during night hours 1-5 and 22-24. In Case 2, the constant WDS energy tariff does not provide sufficient incentives to the W-DSOs to alter the operation of their pumps, resulting to an almost flat power consumption schedule. In Case 3, the W-DSO takes advantage of the low energy tariffs during high solar availability hours 9-17, which results in significantly increasing its electricity consumption during these hours and decreasing it during the rest of the hours.

In Fig. 6, optimizing the energy consumption of WDSs in Cases 1 and 3 has reduced the peak load of the 33-bus distribution system during hours 18-19, as compared to Case 2, where the WDS energy tariff remains constant. In addition, the available WDS energy flexibility has smoothed out the P-DSO active power consumption profile in Case 3, where the W-DSO taps on the available solar generation to operate its pumps during low energy tariff, midday hours.

3) *Operation Schedules of Water Treatment Plants and Storage Tanks*: The optimal freshwater recovery schedules of the water treatment plant, as well as the water flow rate schedules of the tanks for all cases are respectively displayed in Fig. 7 and Fig. 8. As expected, the W-DSOs operate their

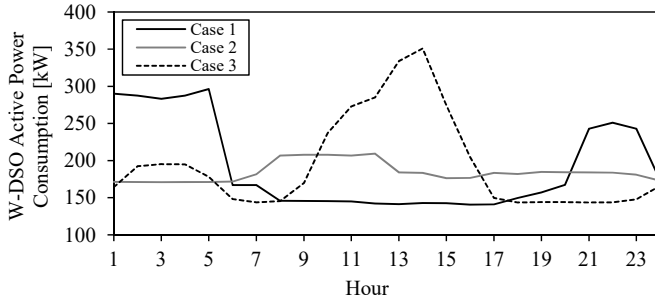


Fig. 5. W-DSO active power consumption schedule

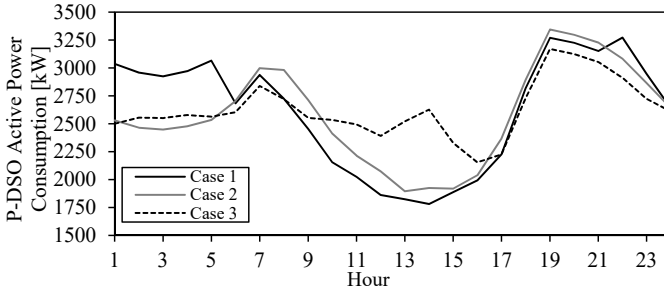


Fig. 6. P-DSO active power consumption schedule

feed pumps to maximize freshwater production during low energy tariff periods which coincide with high solar availability hours in Case 3 (9-17), and night hours in Case 1 (1-5, 22-24). During the same hours, the rest of the pumps operate to store freshwater in the tanks, as shown in Fig. 8, while release the freshwater to supply the freshwater demand of the system when energy tariffs are high. It is worth emphasizing that the energy tariff schemes used in Cases 1 and 3 allow the W-DSO to more efficiently utilize the storage capacity of the tanks as compared to Case 2, by better coordinating the operation of the pumps with water tanks' charge and discharge schedules.

V. CONCLUSION

This paper developed a framework to integrate the energy flexibility of water treatment, desalination and distribution systems in power distribution systems operation. In the proposed framework, the W-DSO co-optimizes the operation of variable speed pumps, water storage tanks, and water treatment plants for minimizing its energy costs, considering different energy tariff schemes. The P-DSO then integrates the available WDS energy flexibility in its day-ahead operation schedule via solving an optimal power flow model. The simulation results demonstrate the capability of the proposed framework in capturing the energy flexibility of WDSs to reduce the energy costs of both power and water distribution systems, while respecting the operational constraints of both systems. The proposed model would reduce the peak electricity demand and smooth out the P-DSO load profile, especially in the case where the operation of WDS is optimized considering the low energy tariffs during periods of high solar generation.

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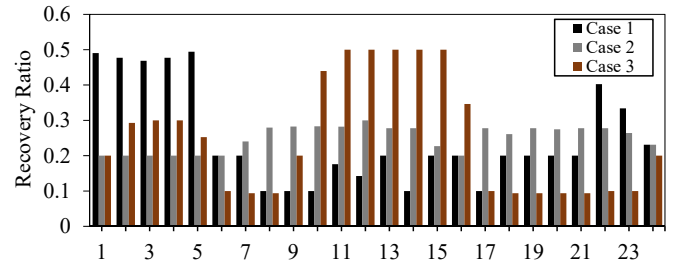


Fig. 7. Water treatment plant operation schedule

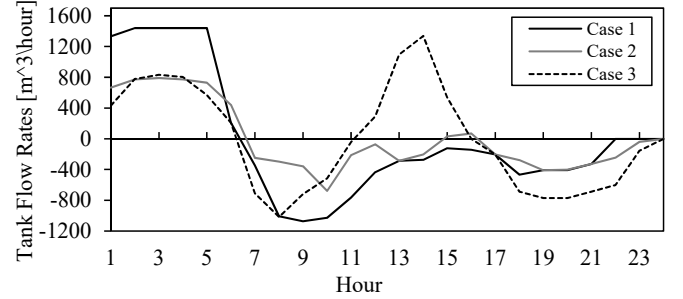


Fig. 8. Water storage tanks operation schedule

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